

Example: Controlling an Elevator

Primitive actions: $up(n)$, $down(n)$, $turnoff(n)$, $open$, $close$.

Fluents: $floor(s) = n$, $on(n, s)$.

Fluent abbreviation: $next_floor(n, s)$.

Action Precondition Axioms:

$Poss(up(n), s) \equiv floor(s) < n.$

$Poss(down(n), s) \equiv floor(s) > n.$

$Poss(open, s) \equiv True.$

$Poss(close, s) \equiv True.$

$Poss(turnoff(n), s) \equiv on(n, s).$

$Poss(no_op, s) \equiv True.$

Elevator Example (cont.)

Successor State Axioms:

$$\begin{aligned} \text{floor}(\text{do}(a, s)) = m \equiv \\ a = \text{up}(m) \vee a = \text{down}(m) \vee \\ \text{floor}(s) = m \wedge \neg \exists n a = \text{up}(n) \wedge \neg \exists n a = \text{down}(n). \end{aligned}$$

$$\begin{aligned} \text{on}(m, \text{do}(a, s)) \equiv \\ a = \text{push}(m) \vee \text{on}(m, s) \wedge a \neq \text{turnoff}(m). \end{aligned}$$

Fluent abbreviation:

$$\begin{aligned} \text{next_floor}(n, s) \stackrel{\text{def}}{=} \text{on}(n, s) \wedge \\ \forall m. \text{on}(m, s) \supset |m - \text{floor}(s)| \geq |n - \text{floor}(s)|. \end{aligned}$$

Elevator Example (cont.)

Golog Procedures:

```
proc serve(n)  
  go_floor(n); turnoff(n); open; close  
endProc
```

```
proc go_floor(n)  
  [floor = n? | up(n) | down(n)]  
endProc
```

```
proc serve_a_floor  
   $\pi n$  [next_floor(n)?; serve(n)]  
endProc
```

Elevator Example (cont.)

Golog Procedures (cont.):

```
proc control  
  while  $\exists n$  on(n) do serve_a_floor endWhile;  
  park  
endProc
```

```
proc park  
  if floor = 0 then open  
  else down(0); open  
  endif  
endProc
```

Elevator Example (cont.)

Initial situation:

$$\text{floor}(S_0) = 4, \quad \text{on}(5, S_0), \quad \text{on}(3, S_0).$$

Querying the theory:

$$\text{Axioms} \models \exists s \text{ Do}(\text{control}, S_0, s).$$

Successful proof might return

$$s = \text{do}(\text{open}, \text{do}(\text{down}(0), \text{do}(\text{close}, \text{do}(\text{open}, \text{do}(\text{turnoff}(5), \text{do}(\text{up}(5), \text{do}(\text{close}, \text{do}(\text{open}, \text{do}(\text{turnoff}(3), \text{do}(\text{down}(3), S_0)))))))))).$$